FM Bandwidth

- Theoretically, FM produces an infinite number of side frequencies,
- In practice, side frequencies with amplitudes < *1%* of unmodulated carrier amplitude are considered insignificant
- Carson's Rule (approximates bandwidth of modulated carrier)

FM Bandwidth../2

- More accurate bandwidth of FM is determined using Bessel Table
- First determine *m* of FM in order to use Bessel table to determine bandwidth

$$B = 2 f_m N$$

  - *N* = No. of side frequency pairs

- Assumptions?

FM Bandwidth../3

Example

- VHF broadcasting uses FM with a maximum frequency deviation of 75 kHz and highest modulation frequency of 15 kHz. Plot the sideband spectrum and calculate the bandwidth required for transmission.

Solution

$$\delta = 75\,kHz \qquad f_m = 15\,kHz \qquad \text{modulation index } m = \frac{\delta}{f_m} = 5$$

Using Bessel table, for $m = 5$, Bessel functions are:

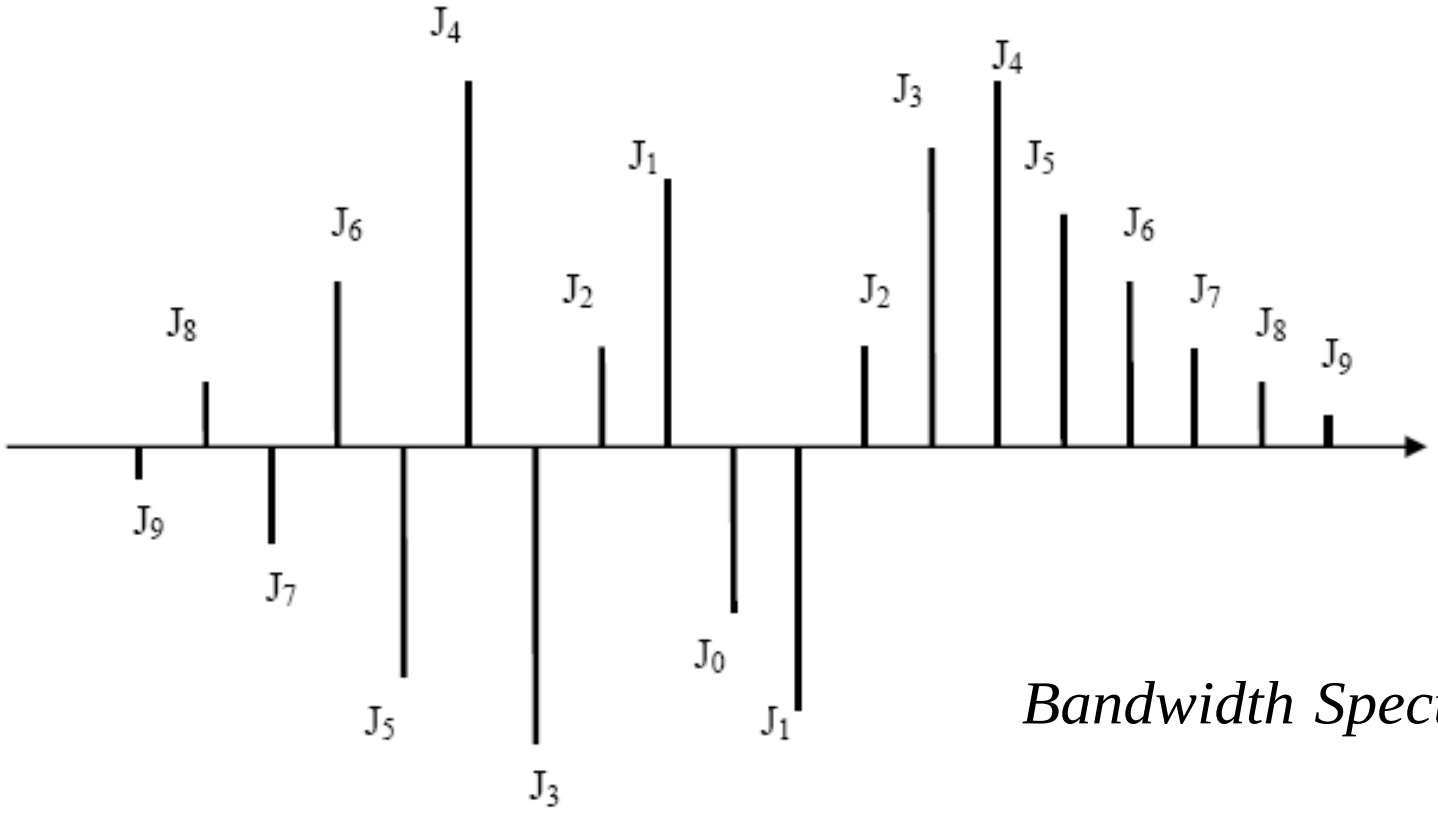
$J_0 = -0.18,\ J_1 = -0.33,\ J_2 = 0.05,\ J_3 = 0.36,\ J_4 = 0.39,\ J_5 = 0.26,\ J_6 = 0.13,$

$J_7 = 0.05,\ J_8 = 0.02,\ J_9 = 0.01$

FM Bandwidth../4

Solution….

- Substituting these in Bessel Equation, the sideband spectrum is as shown below



Bandwidth Spectrum for $m=5$

FM Bandwidth../5

Solution….

$$N = 9 \; sidebands,$$

$$Bandwidth \;\; B = 2 f_m N \;\; = 270\, kHz$$

Using $Carson's\, rule$

$$Bandwidth \;\; B = 2(f_m + \delta) = \; 180\, kHz$$

FM Bandwidth../6

- Modulation index (*m)* can also be determined from Bessel table

- Any FM signal that has only 1 pair of significant side frequencies is ***narrowband FM* (NBFM).**

FM Bandwidth../7

- For NBFM, N = 1

$$Bandwidth \;\; B = \;\; 2\, f_m$$

- Primary purpose of NBFM is to conserve frequency spectrum,
- Any FM signal with a modulation index greater than 0.25 is ***wideband FM***
- Use $f_{m_{max}}$ to determine m and B if complex modulating signal is used.

FM Bandwidth../8

- Bandwidth places limit on number of frequencies that can occupy communication channel

- What happens if FM carrier is over-modulated?